

Coding & Solution

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the implementation quality of the **Rising Waters: A Machine Learning Approach to Flood Prediction** project. The solution successfully integrates machine learning with a Flask web application to provide real-time flood predictions. The project follows clean coding practices, modular design, and fulfills the defined functional requirements.

Instructions:

1. The project code is organized into separate modules for data preprocessing, model training, prediction, and Flask application development.
2. The solution addresses the flood prediction problem using machine learning algorithms and provides accurate prediction results through a web interface.
3. All required source files, trained model files, HTML templates, datasets, and configuration files are included for project execution.
4. The project can be uploaded to a GitHub repository for version control and collaboration.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/KSriMounika/Rising-Waters
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask

Key Features Implemented	• Data preprocessing and visualization • Flood prediction using XGBoost Machine Learning model • User-friendly Flask web application • Real-time flood prediction based on rainfall and weather inputs • Prediction result pages displaying Flood / No Flood status
Pending / Incomplete Features	IBM Cloud deployment, user authentication, real-time weather API integration, database connectivity
Setup / Run Instructions	Install Python dependencies, place the trained model file in the project folder, run app.py using Flask, and open the local server URL in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- Successfully developed a Machine Learning-based flood prediction system.
- Uses historical weather data for accurate prediction.
- Modular architecture and Flask deployment ensure easy maintenance and scalability.
- Future scope: Cloud deployment, live weather data integration, and user authentication.